

Intern Name : Nidhi Rajpurohit

Git – Github Assignment

creating the struture

```
nidhi@nidhi-Predator-PHN16-72:~$ mkdir gha-assignment && cd gha-assignment
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /home/nidhi/gha-assignment/.git/
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git branch -M main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ mkdir -p scripts
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ mkdir -p .github/workflows
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ mkdir -p .github/actions/greet
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ echo "# GitHub Actions Assignment" > README.md
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat README.d
cat: README.d: No such file or directory
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat README.md
# GitHub Actions Assignment
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano index.html
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat index.html
!DOCTYPE html>
<html>
<head>
  <title>GHA Assignment</title>
</head>
<body>
  <h1>Hello from GitHub Actions!</h1>
</body>
</html>
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano scripts/hello.sh
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat scripts/hello.sh
#!/usr/bin/env bash
echo "Hello from GitHub Actions!"
echo "Run number: ${GITHUB_RUN_NUMBER}"
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ chmod +x scripts/hello.sh
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ bash scripts/hello.sh
Hello from GitHub Actions!
Run number:
```

```
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git remote add origin https://github.com/NidhiRajpurohit-64/gha-assignment
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git remote -v
origin  https://github.com/NidhiRajpurohit-64/gha-assignment (fetch)
origin  https://github.com/NidhiRajpurohit-64/gha-assignment (push)
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Initial project setup"
[main (root-commit) a94137a] Initial project setup
 3 files changed, 13 insertions(+)
 create mode 100644 README.md
 create mode 100644 index.html
 create mode 100755 scripts/hello.sh
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/NidhiRajpurohit-64/gha-assignment/'
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push -u origin main
Username for 'https://github.com': NidhiRajpurohit-64
Password for 'https://github.com': 640github.com:
To https://github.com/NidhiRajpurohit-64/gha-assignment
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/NidhiRajpurohit-64/gha-assignment'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git pull origin main --allow-unrelated-histories
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 869 bytes | 869.00 KiB/s, done.
From https://github.com/NidhiRajpurohit-64/gha-assignment
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
hint: You have divergent branches and need to specify how to reconcile them.
hint: You can do so by running one of the following commands sometime before
hint: your next pull:
hint:
hint:     git config pull.rebase false  # merge
hint:     git config pull.rebase true   # rebase
hint:     git config pull.ff only       # fast-forward only
hint:
hint: You can replace "git config" with "git config --global" to set a default
hint: preference for all repositories. You can also pass --rebase, --no-rebase,
hint: or --ff-only on the command line to override the configured default per
hint: invocation.
fatal: Need to specify how to reconcile divergent branches.
```

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hint: You have divergent branches and need to specify how to reconcile them.
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hint:     git config pull.rebase false  # merge
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hint:
hint: You can replace "git config" with "git config --global" to set a default
hint: preference for all repositories. You can also pass --rebase, --no-rebase,
hint: or --ff-only on the command line to override the configured default per
hint: invocation.
fatal: Need to specify how to reconcile divergent branches.
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git pull origin main --no-rebase --allow-unrelated-histories\
> ^C
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git pull origin main --no-rebase --allow-unrelated-histories
From https://github.com/NidhiRajpurohit-64/gha-assignment
 * branch          main      -> FETCH_HEAD
Auto-merging README.md
CONFLICT (add/add): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat > README.md <<'EOF'
# GitHub Actions Assignment

This repository contains my GitHub Actions intermediate assignment.
EOF
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Merge remote repository with local project"
[main 6a60f3a] Merge remote repository with local project
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 28 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 995 bytes | 995.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/NidhiRajpurohit-64/gha-assignment
 f33e965..6a60f3a  main -> main
branch 'main' set up to track 'origin/main'.
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git status
\On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$

```

Question 1

```

nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ tree -a -I .git
.
├── .github
│   ├── actions
│   │   ├── greet
│   │   └── workflows
│   │       └── q1-release-notes.yml
├── index.html
├── README.md
├── scripts
│   └── hello.sh
└──

6 directories, 4 files
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ cat .github/workflows/q1-release-notes.yml
name: Release Notes

on:
  workflow_dispatch:
    inputs:
      version:
        description: Version
        required: true
        default: "1.0.0"
        type: string

      release_type:
        description: Release type
        required: true
        default: patch
        type: choice
        options:
          - patch
          - minor
          - major

jobs:
  prepare:
    runs-on: ubuntu-latest

    outputs:
      build_tag: ${ steps.prepare.outputs.build_tag }
      build_date: ${ steps.prepare.outputs.build_date }

    steps:
      - name: Prepare build information
        id: prepare
        run: |
          BUILD_TAG="v${{ inputs.version }}-${{ inputs.release_type }}-${GITHUB_RUN_NUMBER}"
          BUILD_DATE="$(date +%Y-%m-%d)"

          echo "build tag=$BUILD_TAG" >> "$GITHUB_OUTPUT"

```

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    echo "Build tag: ${ needs.prepare.outputs.build_tag }"
    echo "Build date: ${ needs.prepare.outputs.build_date }"

- name: Write summary
  run: |
    echo "## Release Summary" >> "$GITHUB_STEP_SUMMARY"
    echo "" >> "$GITHUB_STEP_SUMMARY"
    echo "| Field | Value |" >> "$GITHUB_STEP_SUMMARY"
    echo "|---|---|" >> "$GITHUB_STEP_SUMMARY"
    echo "| Version | ${ inputs.version } |" >> "$GITHUB_STEP_SUMMARY"
    echo "| Release Type | ${ inputs.release_type } |" >> "$GITHUB_STEP_SUMMARY"
    echo "| Build Tag | ${ needs.prepare.outputs.build_tag } |" >> "$GITHUB_STEP_SUMMARY"
    echo "| Build Date | ${ needs.prepare.outputs.build_date } |" >> "$GITHUB_STEP_SUMMARY"
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git ststus
git: 'ststus' is not a git command. See 'git --help'.

The most similar command is
  status
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Add Q1 release notes workflow"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 1011 bytes | 1011.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/NidhiRajpurohit-64/gha-assignment
! [remote rejected] main -> main (refusing to allow a Personal Access Token to create or update workflow '.github/workflows/q1-release-notes.yml' without 'workflow' scope)
error: failed to push some refs to 'https://github.com/NidhiRajpurohit-64/gha-assignment'
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 28 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 1011 bytes | 1011.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/NidhiRajpurohit-64/gha-assignment
   6a60f3a..fb17515  main -> main
branch 'main' set up to track 'origin/main'.

```

Question 2

```

nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q2-matrix-build.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .github/workflows/q2-matrix-build.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Add matrix build workflow"
[main af707aa] Add matrix build workflow
 1 file changed, 72 insertions(+)
   create mode 100644 .github/workflows/q2-matrix-build.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.06 KiB | 1.06 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/NidhiRajpurohit-64/gha-assignment
   fb17515..af707aa  main -> main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$

```

Question 3

```
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q3-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Fix Q3 workflow"
[main e449f54] Fix Q3 workflow
 1 file changed, 18 insertions(+)
 create mode 100644 .github/workflows/q3-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Fix Q3 workflow"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 638 bytes | 638.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
To https://github.com/NidhiRajpurohit-64/gha-assignment
   af707aa..e449f54  main -> main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$
```

Question 4

```
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q4-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Everything up-to-date
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Fix Q4 expressions and conditionals"
[main ecf9a8] Fix Q4 expressions and conditionals
 1 file changed, 36 insertions(+)
 create mode 100644 .github/workflows/q4-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 838 bytes | 838.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NidhiRajpurohit-64/gha-assignment
   e449f54..ecfd9a8  main -> main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$
```

Question 5

```
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q5-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Fix Q5 job outputs and dependencies"
[main bc0eb2b] Fix Q5 job outputs and dependencies
 1 file changed, 36 insertions(+)
   create mode 100644 .github/workflows/q5-fix-me.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 815 bytes | 815.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NidhiRajpurohit-64/gha-assignment
   ecf9a8..bc0eb2b  main -> main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$
```

Question 6

```
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git pull
Already up to date.
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/actions/greet/action.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q6-reusable.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ nano .github/workflows/q6-caller.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git add .
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git commit -m "Add composite action and reusable workflow"
[main 05b77f8] Add composite action and reusable workflow
 3 files changed, 80 insertions(+)
   create mode 100644 .github/actions/greet/action.yml
   create mode 100644 .github/workflows/q6-caller.yml
   create mode 100644 .github/workflows/q6-reusable.yml
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 28 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 1.40 KiB | 1.40 MiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NidhiRajpurohit-64/gha-assignment
   bc0eb2b..05b77f8  main -> main
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$ git log --oneline -6
05b77f8 (HEAD -> main, origin/main) Add composite action and reusable workflow
bc0eb2b Fix Q5 job outputs and dependencies
ecfd9a8 Fix Q4 expressions and conditionals
e449f54 Fix Q3 workflow
af707aa Add matrix build workflow
fb17515 Add Q1 release notes workflow
nidhi@nidhi-Predator-PHN16-72:~/gha-assignment$
```